# COMP6008 Reinforcement Learning

## Group Project Planning Document

### Risk-Aware Residential Energy Management under Solar, Demand and EV Uncertainty

Four-week implementation, research and group-work plan

# 1. Project at a Glance

The proposed project investigates whether reinforcement learning (RL) provides a meaningful advantage over a reasonable rule-based household energy controller when decisions must be made sequentially under uncertainty. The household is modelled with rooftop solar PV, a stationary battery, household electricity demand, an electric vehicle (EV), and grid interaction.

The key design decision is to avoid making predictable WA electricity pricing the primary source of complexity. Instead, the research focus is the value of adaptive sequential decision-making when solar generation, household demand and EV availability/requirements vary between episodes.

| Item | Working decision |
|---|---|
| Problem | Residential energy management with PV, battery and EV |
| Main question | Does RL add value beyond a simple heuristic when future energy conditions are uncertain? |
| Environment | 30-minute timesteps; 48 steps per day |
| State | Battery SOC, EV SOC, solar generation, household load, time, EV availability, EV deadline / requirement |
| Action | Six combined battery/EV actions |
| Reward | Grid cost + battery usage/degradation + EV shortfall + peak import penalties |
| Baseline | Rule-based energy controller |
| RL agents | Q-learning, SARSA, Expected SARSA |
| Primary experiments | Baseline comparison, uncertainty levels, hyperparameter tuning, generalisation |
| Submission code/presentation | 23 October 2026, 23:59 |
| Submission report | 30 October 2026, 23:59 |

# 2. Why This Is an RL Problem

A simple rule such as "charge during the day and discharge in the evening" can perform well when conditions are predictable. That is an important criticism and should be incorporated into the project rather than ignored. The project therefore treats the rule-based controller as a serious baseline.

The RL contribution is the possibility of learning when the obvious rule should be modified because current actions change future states. For example, using battery energy now may reduce flexibility later; charging an EV now may compete with household storage; and uncertainty in future solar or demand may make a fixed schedule sub-optimal.

The research should not assume that RL will always win. A useful outcome is to determine the conditions under which RL is competitive with, equal to, or better than the heuristic baseline.

# 3. Proposed Environment and MDP

## 3.1 Household components

- Rooftop PV: produces time-varying generation across the day.
- Household demand: consumes electricity and varies over time.
- Stationary battery: stores surplus energy and supplies demand later.
- EV: introduces a second storage device with availability and a future charging requirement.
- Grid: supplies deficits and can receive surplus energy under the chosen simplified tariff/export assumptions.

## 3.2 State representation

A practical state vector is:

**s_t = [SOC_battery, SOC_EV, solar_t, load_t, time_t, EV_available, EV_deadline]**

| Variable | Meaning | Suggested tabular treatment |
|---|---|---|
| SOC_battery | State of charge of stationary battery | Discretise into bins such as 0-20%, 20-40%, ... 80-100% |
| SOC_EV | State of charge of EV | Use the same or similar bins |
| solar_t | Current PV generation | None / low / medium / high |
| load_t | Current household demand | Low / medium / high |
| time_t | Position within day | 48 half-hour periods |
| EV_available | Whether EV is connected | 0 / 1 |
| EV_deadline | Urgency of EV requirement | Near / medium / far or discretised remaining steps |

### 3.3 Action representation

Use a small combined discrete action space so that tabular RL remains computationally manageable. Let the battery action be -1 (discharge), 0 (idle), or +1 (charge). Let the EV action be 0 (do not charge) or 1 (charge). This creates six possible combined actions.

| Action ID | Battery | EV |
|---|---|---|
| 0 | Idle | Idle |
| 1 | Charge | Idle |
| 2 | Discharge | Idle |
| 3 | Idle | Charge |
| 4 | Charge | Charge |
| 5 | Discharge | Charge |

### 3.4 Reward concept

A candidate reward is a weighted negative-cost formulation:

**R_t = -C_grid - lambda_1 D_battery - lambda_2 P_EV - lambda_3 P_peak**

- C_grid: grid energy cost or equivalent grid-use penalty under the selected modelling assumptions.
- D_battery: penalty representing battery use/degradation.
- P_EV: penalty for not meeting the EV requirement by its deadline.
- P_peak: penalty for undesirable high grid import / peak demand.

The exact numerical weights must be selected and justified by the group using the project literature and sensitivity analysis. Avoid claiming that one weighting is universally correct.

## 4. Uncertainty: The Main Source of Difficulty

The environment should not consist of one perfectly repeatable day. The team should generate multiple scenarios so the agents cannot simply memorise a fixed timetable.

| Scenario | Concept | Purpose |
|---|---|---|
| Low uncertainty | Solar and load close to expected profiles; EV behaviour relatively predictable | Tests whether a simple heuristic is already sufficient |
| Medium uncertainty | Moderate variation in solar and demand; some variation in EV timing | Tests adaptive behaviour |
| High uncertainty | Larger solar/load deviations and more variable EV availability/requirements | Stress-tests policy robustness |

A simple modelling approach is to perturb nominal profiles, for example:

**PV_t' = PV_t (1 + epsilon_PV)**

**Load_t' = Load_t (1 + epsilon_Load)**

The distributions and parameter ranges for these perturbations should be documented and kept consistent between training and testing. Do not use test scenarios to tune the final policies.

## 5. Rule-Based Baseline

Implement the obvious controller before RL. It is a critical component of the scientific design because it answers the central question: what does RL add beyond a reasonable non-learning policy?

1. Use available PV to satisfy household demand first.
2. Use remaining PV to charge the EV and/or battery according to a simple priority rule.
3. When PV is insufficient, use stored battery energy to reduce grid import, subject to SOC constraints.
4. Prioritise satisfying an approaching EV deadline.
5. Use the grid when storage and PV cannot meet the remaining demand.

The exact priority ordering should be frozen before the final comparison so that the baseline is not changed to favour or disadvantage RL after seeing results.

## 6. RL Agents

| Agent | Primary concept | What to compare |
|---|---|---|
| Q-learning | Off-policy TD control | Learning speed, final reward, robustness |
| SARSA | On-policy TD control | Effect of following the behaviour policy during learning |
| Expected SARSA | Expected-value TD control | Whether averaging over next-action probabilities improves stability or performance |

Every agent needs a group-owned implementation rather than simply calling a library training method. Each implementation must have pseudocode, clear mathematical definitions, and an explicit mapping between pseudocode steps and code. The project guidelines also require multiple runs and hyperparameter fine-tuning.

## 7. Experimental Design

### 7.1 Experiment A - Rule-based baseline vs RL

Compare the heuristic controller with Q-learning, SARSA and Expected SARSA on the same environments and evaluation scenarios.

| Metric | Interpretation |
|---|---|
| Average episode reward | Overall RL objective |
| Grid energy imported | Grid dependence |
| Energy cost / grid-use cost | Economic performance under the chosen assumptions |
| EV requirement satisfaction | Whether the future user requirement is met |
| Peak grid import | Ability to avoid concentrated grid demand |
| Battery usage/degradation proxy | Cost of cycling / preserving storage |
| Convergence behaviour | How quickly and consistently an agent learns |

### 7.2 Experiment B - Uncertainty stress test

- Run every controller under low, medium and high uncertainty.
- Use the same evaluation protocol for each controller.
- Compare mean performance and variability across multiple random seeds / runs.

- Look for changes in the relative ranking of the controllers.

### 7.3 Experiment C - Hyperparameter tuning

| Parameter | Example values | Question |
|---|---|---|
| Learning rate alpha | 0.05, 0.10, 0.20 | How quickly should new experience overwrite old estimates? |
| Discount factor gamma | 0.90, 0.95, 0.99 | How strongly should future rewards matter? |
| Initial epsilon | e.g. 1.0 with decay | How much exploration is needed? |
| Epsilon decay | A small set of candidate decay rates | How quickly should exploration decrease? |

Keep the tuning budget modest. The purpose is to demonstrate a scientifically reasoned tuning procedure, not to perform an enormous hyperparameter sweep.

### 7.4 Experiment D - Generalisation to unseen profiles

Train using one collection/distribution of profiles and evaluate on held-out or newly generated profiles. Report whether performance degrades when the policy meets unfamiliar combinations of solar, demand and EV conditions.

## 8. Group Roles and Ownership

The recommended structure is ownership by technical area, not a division in which one student only researches, one only codes, and one only prepares slides. Each member must still understand the whole project.

### Person 1 - Environment and Integration Lead

- Own the MDP formulation and environment specification.
- Implement the household simulator, transition dynamics, battery/EV constraints and grid calculations.
- Implement and validate the rule-based baseline.
- Integrate all three agents into the final training/evaluation pipeline.
- Own final end-to-end reproducibility and notebook integration.

### Person 2 - RL Theory and Agent Lead

- Learn and explain MDPs, Bellman equations, TD learning and exploration/exploitation.
- Implement Q-learning, SARSA and Expected SARSA from the group-approved mathematical formulation.
- Implement epsilon-greedy behaviour and common agent interfaces.
- Own agent hyperparameter configuration and training logic.
- Prepare algorithm pseudocode and the mapping from pseudocode to code.

### Person 3 - Energy Domain and Experimental Analysis Lead

- Research residential energy management, PV, storage, EV charging and uncertainty.
- Investigate and prepare appropriate household/PV/load profiles or synthetic profile generation.
- Own uncertainty scenario design and train/test scenario separation.
- Implement evaluation metrics, plotting and comparative tables.
- Own the first pass of experimental interpretation and limitations.

# 9. Four-Week Execution Plan

## Week 1: October 5-11 - Learn, design and freeze the scope

Primary objective: everyone learns the minimum theory required while the group freezes a feasible project design.

| Day / checkpoint | Person 1 | Person 2 | Person 3 | Whole group |
|---|---|---|---|---|
| Mon-Tue | Draft MDP/environment design | Start RL fundamentals | Start energy-management literature | Agree problem, research question and scope |
| Wed | Write state/action/reward/transition specification | Derive Q-learning/SARSA/Expected SARSA updates | Review energy variables and evaluation metrics | Review each design choice |
| Thu | Build skeleton simulator | Implement agents on a toy environment | Design uncertainty scenarios | Check that simulator assumptions are internally consistent |
| Fri | Add EV and constraints | Connect agent interface to environment | Prepare profiles / synthetic data logic | Freeze v1 state/action/reward design |
| Weekend | Review end-to-end environment | Teach algorithms to team | Teach domain assumptions to team | 30-minute knowledge-transfer session; identify gaps |

### Week 1 definition of done:

- Exact problem statement and research question agreed.
- MDP state, action, reward and transition model documented.
- Three agents confirmed.
- Rule-based baseline concept frozen.
- Uncertainty levels defined conceptually.
- Each member can explain the project architecture.

# 10. Week 2: October 12-18 - Build the complete system

| Workstream | Person 1 | Person 2 | Person 3 | Whole group |
|---|---|---|---|---|
| Environment | Implement Gymnasium-style environment; battery, EV, grid and reward logic | Validate that agent interface matches observation/action spaces | Create/prepare profiles and uncertainty generators | |
| Agents | Integration tests | Finish Q-learning, SARSA, Expected SARSA and epsilon-greedy | Design evaluation calls / result collection | |
| Testing | Constraint checks and transition sanity tests | Toy-environment convergence tests | Check scenario consistency and metric calculations | |
| Mid-week checkpoint | Environment v1 running | All 3 agents can interact with environment | Scenario generation working | Run an end-to-end test |
| Weekend checkpoint | Baseline + simulator stable | All agents train without crashing | First plots/results table | First full experiment |

### Week 2 definition of done:

- Environment runs end-to-end.

- All three agents run on the same environment.
- Rule-based baseline runs.
- Rewards and physical constraints appear sensible.
- First complete training experiment exists.

## 11. Week 3: October 19-23 - Freeze, experiment and submit code/presentation

This is the critical week. Avoid adding major features once experiments start.

| Day | Group target |
|---|---|
| Mon Oct 19 | Freeze environment, reward, action space and baseline. No major redesign after this point. |
| Tue Oct 20 | Run hyperparameter tuning and select final configurations using a documented procedure. |
| Wed Oct 21 | Run uncertainty experiments and multiple seeds/runs. Save raw results. |
| Thu Oct 22 | Generate learning curves, convergence plots, performance tables and final presentation visuals. Record presentation before the deadline if possible. |
| Fri Oct 23 | Submit code + recorded presentation by 23:59. Verify notebook, requirements and reproducibility instructions. |

### Week 3 minimum result package:

- Baseline vs 3 RL agents.
- Low/medium/high uncertainty comparison.
- Hyperparameter tuning evidence.
- Multiple-run summary (mean and variability).
- At least one convergence/learning curve per agent.
- Generalisation evaluation if time permits within the frozen design.

## 12. Week 4: October 24-30 - Report and critical review

| Person | Primary writing/review ownership |
|---|---|
| Person 1 | Environment, MDP, transition dynamics, implementation limitations, reproducibility |
| Person 2 | Agent theory, mathematical formulation, hyperparameters, convergence and algorithm comparison |
| Person 3 | Problem background, energy-management context, uncertainty, results interpretation, limitations |
| Whole group | Critical review, conclusions, improvement proposals, future work, references, final editing and consistency |

The project guidelines require a report with a description/critical review, potential improvements and expansion/future work, plus individual contribution statements. The report is capped at eight A4 pages in IEEE two-column format excluding references, code listings and mathematical derivations.

## 13. Suggested Report Story

1. Introduction / motivation: explain why the energy-management problem matters and why the sequential uncertainty aspect is challenging.

2. Related work: establish residential energy management, RL for energy problems, and relevant algorithm background using academic sources.
3. Problem formulation: define the MDP, state, action, reward and transition model.
4. Method: explain the three agents, baseline, training procedure and hyperparameter tuning.
5. Experiments: define scenarios, random seeds, metrics and train/test separation.
6. Results: compare baseline, Q-learning, SARSA and Expected SARSA.
7. Critical review: discuss what worked, what did not, and whether RL added value relative to the baseline.
8. Improvements: propose evidence-based ways to overcome actual limitations observed in the project.
9. Future directions: propose scope-expanding approaches that were not already used in the project.
10. Conclusions and contributions.

## 14. Part 2 Research Preparation: Improvements and Future Work

This section is especially important because the guidelines require multiple improvements and multiple future directions, supported by academic literature. Hyperparameter tuning does not count as an improvement. For a group of three, plan for at least three substantive improvements and at least three future-work directions, with a minimum of six qualifying academic literature pieces for this report section.

### Potential improvements linked to project weaknesses

| Potential improvement | Possible project limitation it would address | Research needed |
|---|---|---|
| Improve forecasts / uncertainty representation | Current perturbation model may be too simplistic | Forecasting and uncertainty modelling in energy management |
| Improve constraint handling | Penalty-based constraints may allow undesirable temporary violations | Constrained RL / safe RL methods |
| Use richer action representation | Six actions may be too coarse for realistic power scheduling | Discrete-to-continuous control or finer action discretisation |

### Future work that extends the project scope

| Future direction | How it extends the scope |
|---|---|
| Multi-household coordination | Move from one household to coordinated communities / aggregators |
| Multi-agent RL | Allow multiple controllable energy entities to coordinate |
| More advanced RL / hybrid methods | Compare tabular methods with function approximation or hybrid optimisation |

The final choices should be driven by the actual limitations shown in the completed experiments and supported by the group literature review. Do not present these as completed methods unless they are actually implemented.

## 15. Presentation Plan (15-20 minutes)

| Time | Section | Key content |
|---|---|---|
| 0:00-2:00 | Problem | Residential energy management; why the problem matters |
| 2:00-5:00 | Challenge + RL suitability | Sequential decisions, competing objectives, uncertainty; why the rule baseline is important |
| 5:00-9:00 | Environment / MDP | State, action, reward, transition and constraints |

| Time | Section | Key content |
|---|---|---|
| 9:00-12:00 | Agents | Q-learning, SARSA, Expected SARSA and selection rationale |
| 12:00-17:00 | Results | Learning curves, baseline comparison, uncertainty results, convergence / stability |
| 17:00-19:00 | Critical review | When RL helps, when it does not, and important limitations |
| 19:00-20:00 | Conclusion | Main finding and future direction |

## 16. Final Notebook Structure

1. Introduction and project objective
2. Environment and MDP formulation
3. Data/profile preparation and uncertainty generation
4. Rule-based baseline
5. Q-learning implementation
6. SARSA implementation
7. Expected SARSA implementation
8. Training and evaluation framework
9. Hyperparameter tuning
10. Experimental scenarios
11. Results and visualisation
12. Statistical / comparative analysis
13. Limitations and interpretation
14. Reproducibility and setup instructions

## 17. Suggested Project Folder Structure

**RL-Energy-Management/**

```
├── notebook/
│   └── final_project.ipynb
├── src/
│   ├── environment.py
│   ├── q_learning.py
│   ├── sarsa.py
│   ├── expected_sarsa.py
│   ├── baseline.py
│   └── evaluation.py
├── data/
│   ├── raw/
│   └── processed/
├── experiments/
│   ├── configs/
│   └── results/
├── figures/
├── requirements.txt
└── README.md
```

## 18. Group Coordination and Knowledge Transfer

Because the course guidelines state that all group members are expected to understand every part of the submitted work, weekly ownership hand-offs are essential.

| Meeting | Agenda | Output |
|---|---|---|
| Weekly kickoff (30-45 min) | Review goals, blockers and assignments | Three concrete targets for the week |
| Mid-week technical check (20-30 min) | Show working code / research progress | Issues assigned to an owner |
| Weekend knowledge transfer (30-45 min) | Each member explains their area; other two ask questions | Shared understanding and unresolved questions |
| Week 3 results review | Inspect raw results before writing conclusions | Agreed interpretation and plots |
| Final rehearsal | Random questioning and timing practice | Every member can defend the project |

### Questions the team should be able to answer

- Why is this actually a sequential decision problem?
- Why is a rule-based controller a necessary baseline?
- What makes the future uncertain in our environment?
- Why are these state variables sufficient?
- Why did we choose six discrete actions?
- How does the reward encode our objectives?
- How does Q-learning differ from SARSA and Expected SARSA?
- What does alpha do? What does gamma do? What does epsilon do?
- What happens if the EV leaves before its target SOC is reached?
- What constraints are enforced by the environment rather than by the reward?
- How did we prevent test data from influencing training/tuning?
- Under what conditions did RL outperform the baseline?
- What are the strongest limitations of the current study?

## 19. Contribution Tracking

Keep evidence from the first day. The guidelines require contribution attribution and percentages, and the notebook/report should identify contributors by section.

| Date | Member | Task | Evidence / location |
|---|---|---|---|
| | Person 1 | | |
| | Person 2 | | |
| | Person 3 | | |
| | | | |
| | | | |

Recommended practice: use version control commits or a dated task log, keep research notes with references, and add an author/contributor label to relevant notebook and report sections. Do not fabricate contribution records after the fact.

## 20. Scope Control and Risk Management

| Risk | Mitigation |
|---|---|
| Environment becomes too complex | Freeze to one household, 30-minute steps, three agents and a small discrete action space. |

| Risk | Mitigation |
|---|---|
| RL does not outperform baseline | Treat this as a meaningful result; analyse conditions where the baseline is already strong. |
| Tabular state space explodes | Reduce/discretise state variables before adding complexity; avoid unnecessary state dimensions. |
| Agents learn poorly | Check reward scaling, exploration, learning rate, episode length and transition correctness before adding algorithms. |
| No time for a large real dataset | Use a documented profile-generation or public-profile approach that can be implemented reliably. |
| Presentation becomes too long | Rehearse to about 18-19 minutes and prioritise environment, agents and results. |
| Group members cannot explain other sections | Run weekly knowledge-transfer sessions and final random-question rehearsal. |

## 21. Final Submission Checklist

- Environment implements observation/state, reward, actions and transition logic.
- Three RL agents are implemented by the group and match documented pseudocode.
- A rule-based baseline is included and kept unchanged for final comparison.
- Multiple runs / random seeds are used.
- Hyperparameter fine-tuning is demonstrated.
- Learning curves / convergence or comparable training visuals are included.
- Performance tables include operational metrics, not reward alone.
- Uncertainty experiments are included.
- Unseen-profile generalisation is assessed if the frozen scope allows it.
- Notebook is organised with clear markdown explanations and comments.
- Setup and reproduction instructions are complete; requirements.txt included if required.
- Presentation is no longer than 20 minutes.
- Report uses IEEE format and stays within the eight-page limit excluding allowed appendices/references.
- Part 2 includes the required number of improvements and future directions and at least six academic literature pieces.
- Contribution statements are complete and consistent with the actual work.
- Every group member can explain every part of the submission.
- Blackboard has been checked for the latest project guideline version before final submission.

## 22. One-Sentence Project Story

**We test whether RL can learn adaptive household energy-storage decisions that remain competitive with a simple rule-based controller when solar generation, demand and EV requirements become uncertain.**

*Planning basis: COMP6008 Reinforcement Learning - Group Project, Presentation, and Report Guidelines, Curtin University, version dated July 29, 2026, as provided in the conversation. The guidelines state that the specification may be updated on Blackboard; the group should verify the latest version before submission.*